

Matthew Douglas

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EDUCATION

Queens College, City University of New York

Flushing, NY

Bachelor of Science, Computer Science | Major GPA: 3.52/4.0

Jan 2022 – Jun 2027

Relevant Coursework: Data Structures, Object-Oriented Programming, Computer Organization and Assembly Language, Database Systems, Theory of Computation, Discrete Structures, Software Engineering, Deep Learning and Systems, Operating Systems, Design and Analysis of Algorithms, Internet and Web Technologies

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, HTML/CSS, SQL

Frameworks & Libraries: Node.js, Express, EJS, Mongoose, SFML, Flask, NumPy, Matplotlib

Databases & Tools: MongoDB, MySQL, Docker, Git/GitHub, npm, Linux, Fusion360

Certifications: Google IT Support Certificate, NOCTI – Robotics Engineering, OSHA Certified

PROJECTS

Study Buddy — AI-Powered Study Companion

Jul 2026 – Aug 2026

Node.js, Express, EJS, MongoDB, Google Gemini API

- Built a full-stack web app that generates AI-powered summaries and auto-graded multiple-choice quizzes from user-submitted notes using Google's Gemini API.
- Implemented session-based authentication with bcrypt password hashing, MongoDB-backed session storage, and authenticated middleware protecting user routes.
- Designed a Mongoose data layer (Users, Sessions) and RESTful Express routes following an MVC architecture with server-rendered EJS views.

Intermediate Code Generator

Nov 2025 – Dec 2025

C++ (no STL, no external libraries)

- Built a compiler front-end in pure C++ that lexes and parses arithmetic expressions into an abstract syntax tree (AST), with a hand-written symbol table.
- Designed a custom bytecode instruction set and implemented a code-generation pass that traverses the AST to emit executable bytecode.

Sorting Algorithm Visualizer

Apr 2026

C++, SFML

- Developed an interactive C++ application using the SFML graphics library to visualize 8 sorting algorithms in real time: Bubble, Selection, Insertion, Shell, Merge, Quick, Heap, and Radix Sort.
- Implemented Fisher-Yates shuffle to randomize input arrays and built an interactive selection UI for comparing algorithm behavior side by side.

Personal Portfolio Website

May 2026 – Present

HTML, CSS, JavaScript

- Designed and built a responsive single-page portfolio using vanilla HTML, CSS, and JavaScript with CSS Grid/Flexbox layouts and a mobile breakpoint.
- Implemented a typewriter-effect hero animation and a scroll-synced work-experience timeline using custom JavaScript, with no external frameworks.

EXPERIENCE

Target

Brooklyn, NY

Technology Sales Consultant

Apr 2023 – Feb 2024

- Advised customers on consumer electronics and technology products, delivering personalized recommendations that drove department sales performance.

NVSN

Remote

Production Financial Coordinator

Jun 2022 – Jan 2023

- Tracked and reconciled production expenditures against budget in Microsoft Excel, flagging and resolving discrepancies to keep projects on track.

ACTIVITIES & LEADERSHIP

FRC Robotics Team, Thomas A. Edison High School

Jamaica, NY

Head Programmer & CAD Designer

2019 – 2020

- Led the programming team in developing autonomous and tele-operated control systems for the competition robot.

VEX Robotics Team, Thomas A. Edison High School

Jamaica, NY

Co-Captain & Head Programmer

2016 – 2020

- Supervised and mentored team members across engineering, electrical, and programming disciplines as co-captain.